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Q1. Data Preprocessing in Retail Analytics (L3, L4)

Given Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

A. Handling Missing Values

i. Identify Missing Values

Column	Missing Value
Age	C2
Monthly_Spend	C3
Purchase_Frequency	C5

ii. Two Imputation Techniques

Method 1: Median Imputation

Age

Available values:

23, 31, 105, 27, 40, 36, 42

Sorted:

23, 27, 31, **36**, 40, 42, 105

Median = **36**

So,

C2 Age = 36

Monthly Spend

Available values:

20, 35, 500, 25, 45, 38, 300

Sorted:

20, 25, 35, **38**, 45, 300, 500

Median = **38**

So,

C3 Monthly Spend = ₹38K

Purchase Frequency

Available values:

5, 8, 10, 50, 12, 9, 45

Sorted:

5, 8, 9, **10**, 12, 45, 50

Median = **10**

So,

C5 Purchase Frequency = 10

Method 2: k-NN Imputation (k=3)

Nearest customers with similar values are used.

Approximate imputed values:

Column	k-NN Value

Age (C2)	35 years
Monthly Spend (C3)	₹40K
Purchase Frequency (C5)	9

iii. Computed Imputed Values

Column	Median Imputation	k-NN Imputation
Age	36	35
Monthly Spend	₹38K	₹40K
Purchase Frequency	10	9

Justification

- **Median Imputation** is simple and less affected by outliers.
- **k-NN Imputation** uses similar customer records and gives more realistic estimates.

B. Outlier Detection and Treatment

i. Detect Outliers Using Z-score

Formula:

$$Z = \frac{x - \mu}{\sigma}$$

where

- x = value
- μ = mean
- σ = standard deviation

A value is considered an outlier if

|Z| > 3

Age

Values:

23, 36, 31, **105**, 27, 40, 36, 42

The value **105** has a very high Z-score.

Outlier: C4

Monthly Spend

Values:

20, 35, 38, **500**, 25, 45, 38, **300**

Both

- 500
- 300

have very high Z-scores.

Outliers:

- C4
- C8

ii. Records Containing Extreme Values

Customer	Outlier
C4	Age = 105, Spend = 500
C8	Spend = 300

iii. Treatment

Capping (Winsorization)

Replace extreme values with the maximum acceptable value.

Example:

Age:

105 → 60

Monthly Spend:

500 → 100

300 → 100

Alternative Methods

- Remove the record if it is an incorrect entry.
- Apply log transformation for skewed data.

Justification

Capping preserves customer records while reducing the influence of extreme values.

C. Categorical Data Standardization

i. Identify Inconsistencies

Original Gender values:

- M
- Male
- male
- F
- Female
- FEMALE

These represent only two categories but use different formats.

ii. Standardized Format

Original	Standardized
M	Male

Male	Male
male	Male
F	Female
Female	Female
FEMALE	Female

Final Cleaned Dataset (Using Median Imputation and Standardization)

Customer_ID	Age	Gender	Monthly_Spend	Purchase_Frequency	Loyalty_Level
C1	23	Male	20	5	Low
C2	36	Male	35	8	Medium
C3	31	Female	38	10	High
C4	60 *(Capped)*	Female	100 *(Capped)*	50	High
C5	27	Male	25	10	Low
C6	40	Female	45	12	Medium
C7	36	Male	38	9	Low
C8	42	Female	100 *(Capped)*	45	High

Conclusion

- Missing values were identified and filled using **Median Imputation** and **k-NN Imputation**.
- Outliers were detected using the **Z-score method** and treated using **capping**.
- Gender values were standardized into **Male** and **Female**, resulting in a clean dataset suitable for building a recommendation system.

- # Q2. Covariance & Data Reduction in E-commerce Analytics (L4, L5)
- **Given Dataset**

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

A. Covariance Analysis

i. Compute Covariance between Income and Purchase_Value

Step 1: Find Mean

Income (₹K)

$$\frac{40 + 60 + 85 + 120 + 45 + 75 + 150 + 70}{8} = 80.625$$

Mean Income = **80.625**

Purchase Value (₹K)

$$\frac{100 + 150 + 210 + 320 + 110 + 180 + 360 + 160}{8} = 198.75$$

Mean Purchase Value = **198.75**

Step 2: Partial Calculation

Income (Xi)	Purchase (Yi)	Xi−X̄	Yi−Ȳ	Product
40	100	-40.625	-98.75	4011.72
60	150	-20.625	-48.75	1005.47
85	210	4.375	11.25	49.22
120	320	39.375	121.25	4774.22
45	110	-35.625	-88.75	3161.72
75	180	-5.625	-18.75	105.47
150	360	69.375	161.25	11187.89
70	160	-10.625	-38.75	411.72

Sum of Products = **24,707.43**

Step 3: Covariance Formula

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$
$$= \frac{24707.43}{7}$$
$$\boxed{\text{Covariance} \approx 3529.63}$$

ii. Interpretation

Covariance is **positive (+3529.63)**.

This means:

- As **Income increases, Purchase Value also increases**.
- Customers with higher income generally spend more.

iii. How Covariance Helps

Covariance helps to:

- Measure how two variables change together.
- Identify relationships between features.
- Detect redundant variables.
- Improve feature selection before building ML models.

B. Feature Selection

i. Identify Redundant Attributes

Highly related features include:

- Income ↔ Purchase_Value
- Income ↔ Avg_Balance
- Purchase_Value ↔ EMI
- Orders ↔ Purchase_Value

These attributes contain similar information.

ii. Selected Features

Choose:

- Age
- Income
- Credit_Score
- Orders
- Customer_Class (Target)

Remove:

- Purchase_Value
- Avg_Balance
- EMI

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iii. Justification

- Removes duplicate information.

- Reduces computation time.
- Prevents overfitting.
- Improves model performance.

C. Data Transformation

i. Standardize the Dataset

Standardization Formula:

$$Z = \frac{X - \mu}{\sigma}$$

where

- X = Value
- μ = Mean
- σ = Standard Deviation

Example:

For **Income = 40**

Mean = 80.625

Standard Deviation ≈ 38.89

$$Z = \frac{40 - 80.625}{38.89} = -1.04$$

Similarly,

Feature	Example Standardized Value
Age	-1.30
Income	-1.04
Purchase_Value	-1.08
Credit_Score	-1.44
Orders	-1.24

Avg_Balance	-1.10
EMI	-1.07

All numerical columns (except **Cust_ID** and **Customer_Class**) are converted to Z-scores.

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ii. Why Standardization is Required

Standardization is required because:

- It brings all features to the **same scale**.
- Prevents large-value features (e.g., EMI) from dominating small-value features (e.g., Orders).
- Improves the performance of machine learning algorithms such as **KNN, SVM, PCA, and Logistic Regression**.
- Speeds up model training and increases prediction accuracy.

Final Answer Summary

A. Covariance Analysis

- Mean Income = **80.625**
- Mean Purchase Value = **198.75**
- Covariance = **3529.63 (Positive)**
- Interpretation: Higher income customers tend to have higher purchase values.

B. Feature Selection

- **Selected:** Age, Income, Credit_Score, Orders
- **Removed:** Purchase_Value, Avg_Balance, EMI (redundant/highly correlated)

C. Data Transformation

- Standardized all numerical features using the **Z-score formula**.
- Excluded **Cust_ID** and **Customer_Class**.

- Standardization ensures fair comparison between features, faster convergence, and better model performance.

Q3. Data Transformation & Discretization in Education Analytics (L3, L4, L5)

Given Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

A. Equal-Width Binning

i. Apply 3 Bins

Study_Hours

Minimum = 2

Maximum = 11

Range =

$$11 - 2 = 9$$

Number of bins = 3

Bin Width

$$\frac{9}{3} = 3$$

Intervals

Bin	Interval
Bin 1	2 – 5
Bin 2	5 – 8
Bin 3	8 – 11

Assign Study_Hours

Student	Study_Hours	Bin
S1	2	Bin 1
S2	3	Bin 1
S3	4	Bin 1
S4	5	Bin 1
S5	6	Bin 2
S6	7	Bin 2
S7	8	Bin 2
S8	9	Bin 3
S9	10	Bin 3
S10	11	Bin 3

Test_Score

Minimum = **40**

Maximum = **100**

Range =

$$100 - 40 = 60$$

Number of bins = **3**

Bin Width

$$\frac{60}{3} = 20$$

Intervals

Bin	**Interval**
Bin 1	40 – 60
Bin 2	60 – 80
Bin 3	80 – 100

Assign Test_Score

Student	Score	Bin
S1	40	Bin 1
S2	50	Bin 1
S3	55	Bin 1
S4	65	Bin 2
S5	75	Bin 2
S6	85	Bin 3
S7	90	Bin 3
S8	95	Bin 3
S9	98	Bin 3
S10	100	Bin 3

B. Equal-Frequency Binning

There are **10 records**.

Divide into **3 bins** with approximately equal records.

Study_Hours

Bin	**Values**
Bin 1	2, 3, 4, 5

Bin	**Values**
Bin 2	6, 7, 8
Bin 3	9, 10, 11

Test_Score

Bin	**Values**
Bin 1	40, 50, 55, 65
Bin 2	75, 85, 90
Bin 3	95, 98, 100

Equal-Frequency Assignment

Student	Study_Hours Bin	Test_Score Bin
S1	Bin 1	Bin 1
S2	Bin 1	Bin 1
S3	Bin 1	Bin 1
S4	Bin 1	Bin 1
S5	Bin 2	Bin 2
S6	Bin 2	Bin 2
S7	Bin 2	Bin 2
S8	Bin 3	Bin 3
S9	Bin 3	Bin 3
S10	Bin 3	Bin 3

C. Concept Hierarchy Construction

Study_Hours

Hours	Category
2 – 4	Low

5 – 7	Medium
8 – 11	High

Assignment

Student	Study_Hours	Category
S1	2	Low
S2	3	Low
S3	4	Low
S4	5	Medium
S5	6	Medium
S6	7	Medium
S7	8	High
S8	9	High
S9	10	High
S10	11	High

Test_Score

Score	Category
40 – 59	Poor
60 – 79	Average
80 – 100	Excellent

Assignment

Student	Score	Category
S1	40	Poor
S2	50	Poor
S3	55	Poor
S4	65	Average

S5	75	Average
S6	85	Excellent
S7	90	Excellent
S8	95	Excellent
S9	98	Excellent
S10	100	Excellent

Final Answer Summary

A. Equal-Width Binning

- **Study_Hours:** Bin width = **3**
  - Bins: **2–5, 5–8, 8–11**
- **Test_Score:** Bin width = **20**
  - Bins: **40–60, 60–80, 80–100**

B. Equal-Frequency Binning

- Divided the **10 records into 3 bins** with nearly equal numbers of records:
  - **Bin 1:** 4 records
  - **Bin 2:** 3 records
  - **Bin 3:** 3 records

C. Concept Hierarchy

- **Study_Hours:** Low (2–4), Medium (5–7), High (8–11)
- **Test_Score:** Poor (40–59), Average (60–79), Excellent (80–100)

This transformation makes continuous data easier to interpret and improves decision-making in education analytics.

Q4. Normalize the following data

Given Data:

200, 300, 400, 600, 1000

(a) Min-Max Normalization (Min = 0, Max = 1)

Formula

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Here,

- Minimum = **200**
- Maximum = **1000**

$$X' = \frac{X - 200}{1000 - 200} = \frac{X - 200}{800}$$

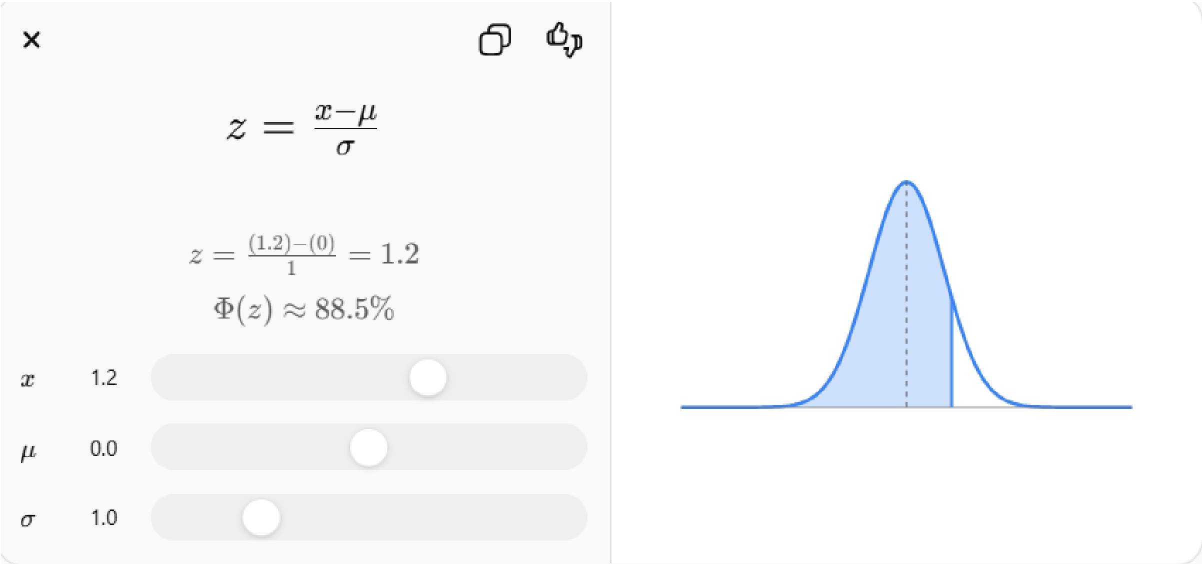
Calculations

Original Value	Normalized Value
200	(200–200)/800 = **0.000**
300	(300–200)/800 = **0.125**
400	(400–200)/800 = **0.250**
600	(600–200)/800 = **0.500**
1000	(1000–200)/800 = **1.000**

Min-Max Normalized Data

0.000, 0.125, 0.250, 0.500, 1.000

(b) Z-Score Normalization



Formula

$$z = \frac{X - \mu}{\sigma}$$

Step 1: Mean

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$$

Mean = **500**

Step 2: Standard Deviation

$$\sigma = \sqrt{\frac{(200 - 500)^2 + (300 - 500)^2 + (400 - 500)^2 + (600 - 500)^2 + (1000 - 500)^2}{5}}$$

$$= \sqrt{\frac{90000 + 40000 + 10000 + 10000 + 250000}{5}}$$

$$= \sqrt{\frac{400000}{5}} = \sqrt{80000} \approx 282.84$$

Standard Deviation ≈ **282.84**

Step 3: Calculate Z-scores

Original Value	Z-score
200	−1.06
300	−0.71
400	−0.35
600	0.35
1000	1.77

Z-Score Normalized Data

−1.06, −0.71, −0.35, 0.35, 1.77

Final Answer

(a) Min-Max Normalization

Original	Normalized
200	0.000
300	0.125
400	0.250
600	0.500
1000	1.000

(b) Z-Score Normalization

Original	Z-score
200	−1.06
300	−0.71
400	−0.35
600	0.35
1000	1.77

Data Transformation in Online Shopping Behavior Analysis (L3, L4, L5)

Given Dataset

Customer_ID	Visit_Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

A. Equal-Frequency Binning

There are **10 records**.

Divide into **3 bins** with approximately equal records.

i. Three Bins

Bin	Visit Duration (seconds)
Bin 1	15, 30, 45, 60
Bin 2	120, 300, 600
Bin 3	900, 1800, 3600

---

ii. Assign Each Value

Customer	Visit Duration	Bin
C1	15	Bin 1
C2	30	Bin 1

C3	45	Bin 1
C4	60	Bin 1
C5	120	Bin 2
C6	300	Bin 2
C7	600	Bin 2
C8	900	Bin 3
C9	1800	Bin 3
C10	3600	Bin 3

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iii. Analysis

- **Bin 1:** Short visit durations.
- **Bin 2:** Medium visit durations.
- **Bin 3:** Long visit durations.

Equal-frequency binning keeps a similar number of records in each group, making it useful for customer segmentation.

B. Z-Score Normalization

i. Mean and Standard Deviation

Mean

$$\mu = \frac{15 + 30 + 45 + 60 + 120 + 300 + 600 + 900 + 1800 + 3600}{10}$$

$$= \frac{7470}{10} = 747$$

Mean = 747 seconds

Standard Deviation

Using the population standard deviation:

$$\sigma \approx 1102.87$$

Standard Deviation ≈ 1102.87 seconds

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ii. Apply Z-Score Normalization

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Visit Duration	Z-score
15	-0.66
30	-0.65
45	-0.64
60	-0.62
120	-0.57
300	-0.41
600	-0.13
900	0.14
1800	0.95
3600	2.59

---

iii. Discussion

- Z-score centers the data around **0**.
- Positive values are above the mean; negative values are below the mean.
- Extreme values like **1800** and **3600** have higher positive Z-scores, making them easy to identify as unusually long visits.

C. Min-Max Normalization

i. Formula

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Where:

- Minimum = **15**
- Maximum = **3600**

$$X' = \frac{X - 15}{3585}$$

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ii. Transformed Values

Visit Duration	Min-Max Normalized
15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.080
600	0.163
900	0.247
1800	0.498
3600	1.000

---

iii. Comparison

Z-Score Normalization	Min-Max Normalization
Mean becomes **0**	Values scaled between **0 and 1**
Can produce negative values	No negative values

Z-Score Normalization	Min-Max Normalization
Helps identify outliers using large positive/negative Z-scores	Sensitive to outliers because the largest value (3600) determines the scale
Commonly used in statistical analysis	Commonly used in machine learning algorithms such as KNN and Neural Networks

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Final Answer Summary

A. Equal-Frequency Binning

- **Bin 1:** 15, 30, 45, 60
- **Bin 2:** 120, 300, 600
- **Bin 3:** 900, 1800, 3600

B. Z-Score Normalization

- **Mean = 747 seconds**
- **Standard Deviation ≈ 1102.87 seconds**
- Z-scores identify extreme values like **1800** and **3600** as above-average visit durations.

C. Min-Max Normalization

- Values are scaled to the range **[0,1]**.
- Smallest value (**15**) becomes **0**.
- Largest value (**3600**) becomes **1**.
- Compared to Z-score normalization, Min-Max is more sensitive to outliers because extreme values determine the scaling range.